Technique Packet No. 3

The Multiplication Principle — how to count without counting.

Name: ______________________ Date: ______________________

Sam is getting dressed. He has 3 clean shirts — red, blue, green — and 2 pairs of shorts — black and tan. How many different outfits can he put together?

You could try to list them all and hope you don't miss any. Or you could notice something. For each of his 3 shirts, he has 2 choices of shorts — so the outfits come in 3 groups of 2. That is 3 × 2 = 6, and you didn't have to list a single one. This is the whole trick: **when you build something by making a choice at each stage, you don't count the results one by one — you multiply the number of choices at each stage.**

	black	*tan*
red		
blue		
green		

3 × 2 = 6

Every shirt (a row) can meet every pair of shorts (a column). 3 rows × 2 columns = 6 boxes — six outfits.

THE TECHNIQUE — The Multiplication Principle.
When you build something by making a choice at each stage:

1. Break the job into stages — first this, then this, then this.
2. Count how many choices you have at each stage.
3. Multiply those numbers together. That product is the total.

When to reach for it: any "how many different ___ can you make, arrange, or choose" question that you build step by step — outfits, meals, codes, license plates, routes. Don't list them all — multiply.

Watch how it works

WORKED EXAMPLE 1

Sam has 3 shirts and 2 pairs of shorts. How many different outfits can he make?

1. There are two stages: first pick a shirt, then pick a pair of shorts.

2. Stage 1 — the shirt — has 3 choices. Stage 2 — the shorts — has 2 choices.

3. Multiply the stages: $3 \times 2 =$ **6 outfits**.

4. (The grid above shows all six. Notice you never had to list them to know there were six.)

WORKED EXAMPLE 2 — more stages, where listing would be painful

A kids' menu lets you pick 1 of 4 main dishes, then 1 of 3 drinks, then 1 of 2 desserts. How many different meals are possible?

1. Three stages this time: main, then drink, then dessert.

2. Count the choices at each: 4 mains, 3 drinks, 2 desserts.

3. Multiply all three stages: $4 \times 3 \times 2 =$ **24 meals**.

4. Listing all 24 would fill the page; multiplying the stages takes one line. That is exactly when this technique earns its keep.

Watch how it works (one more)

WORKED EXAMPLE 3 — when every stage has the same choices

A treasure chest opens with a 3-symbol code. Each symbol can be any one of 5 shapes, and shapes may repeat. How many different codes are there?

1. Three stages: the first symbol, the second symbol, the third symbol.

2. Each stage has the same 5 choices — and because shapes may repeat, picking one does not use it up. The choices don't run out.

3. So every stage still has 5 choices: $5 \times 5 \times 5$ = **125 codes**.

4. Watch for that word "repeat." If shapes could *not* repeat, the stages would shrink to $5 \times 4 \times 3$ — but here they don't.

All three are the same move: split the job into stages, count the choices at each stage, and multiply.

Now you

Show the stages and the multiplication — not just the final number.

1. An ice-cream stand has 4 flavors and 3 kinds of cone. How many different single-scoop cones can you order?

Count the stages — how many choices at each step:

$\square \times \square \times \square \times \square =$ ______________

(use only as many boxes as you have stages)

2. For breakfast you choose 1 of 3 cereals, then 1 of 2 fruits, then 1 of 3 juices. How many different breakfasts are possible?

Count the stages — how many choices at each step:

$\square \times \square \times \square \times \square =$ ______________

(use only as many boxes as you have stages)

3. A 3-digit code uses the digits 1, 2, 3, 4, 5, and digits may repeat. How many different codes are there?

Count the stages — how many choices at each step:

$\square \times \square \times \square \times \square =$ ______________

(use only as many boxes as you have stages)

4. At a sandwich shop you pick 1 of 3 breads, 1 of 4 fillings, 1 of 2 sauces, and then decide toasted or not toasted. How many different sandwiches are possible?

Count the stages — how many choices at each step:

$\boxed{} \times \boxed{} \times \boxed{} \times \boxed{} =$ __________

(use only as many boxes as you have stages)

5. To get to school, Mia first walks to the park by one of 2 paths, then walks from the park to school by one of 3 paths. How many different ways can she get to school?

Count the stages — how many choices at each step:

$\boxed{} \times \boxed{} \times \boxed{} \times \boxed{} =$ __________

(use only as many boxes as you have stages)

6. How many two-digit numbers have an even tens digit and an odd ones digit? (Think of choosing the tens digit, then the ones digit.)

Count the stages — how many choices at each step:

$\boxed{} \times \boxed{} \times \boxed{} \times \boxed{} =$ __________

(use only as many boxes as you have stages)

Answer Key — for parents

What this packet teaches. One named technique — when you build something by a choice at each stage, multiply the number of choices at each stage — shown first (the outfit hook, the grid picture, and three worked examples), then applied six times. The three worked examples are deliberately different *kinds* of the same move: two independent stages (1), several stages where listing would be hopeless (2), and stages that all share the same choices with repeats allowed (3) — so every kind of problem below has been modeled before they meet it. Each problem has a “stages” box printed under it: the goal is to see the factors, not just the total.

Answers.

1. Two stages: 4 flavors × 3 cones = **12**. (Echoes Worked Example 1.)
2. Three stages: 3 × 2 × 3 = **18**. (Echoes Worked Example 2.)
3. Each of the 3 digits has 5 choices and they may repeat: 5 × 5 × 5 = **125**. (Echoes Worked Example 3 — watch they don’t shrink it to 5 × 4 × 3.)
4. Four stages, and “toasted or not” is just a stage with 2 choices: 3 × 4 × 2 × 2 = **48**.
5. It doesn’t say “how many ways,” but it is the same idea: 2 paths to the park, then 3 to school, 2 × 3 = **6 ways**.
6. Choosing a two-digit number is two stages. The tens digit must be even *and* can’t be 0 (or it wouldn’t be a two-digit number), so it is 2, 4, 6, 8 — 4 choices. The ones digit is odd: 1, 3, 5, 7, 9 — 5 choices. So 4 × 5 = **20**.

What to watch for. Three slips. First, **adding instead of multiplying** (“4 and 3, that’s 7”) — point back to the grid: each of the 4 flavors pairs with *all* 3 cones, so it’s 4 groups of 3. Second, **shrinking the choices when they shouldn’t** — on problem 3 the digits repeat, so every stage stays at 5; only drop the count when a thing gets used up. Third, the old habit of writing just the final number: the “stages” box is there to be filled — ask to see the factors and the × signs, the way the worked examples show them. Problems 5 and 6 are the stretch: 5 hides the stages inside a journey, and 6 hides a sneaky restriction (the tens digit can’t be 0) inside an ordinary-looking question.

How to run it. Read the hook, the grid picture, and all three worked examples out loud together before they touch problem 1. The problems they do solo, in one sitting or two. This is a learning packet, not a test: helping on a stuck problem is fine, but help by pointing back at the matching worked example — and at the stages box — not at the answer.